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BUSINESS INTELLIGENCE

PHASE 2

LECTURER : DR NOR AZIZAH ALI

Impact of Severe Weather Events on US Commercial Flight Operations:
A 2016 Big Data Analytics Approach

NO	NAME	MATRIC NO
1.	PRAVINRAJ A/L SIVABATHI	A23CS0171
2.	DHESHIEGHAN A/L SARAVANA MOORTHY	A23CS0072
3.	NEO LI XIN	A23CS0253

1.0 BI Architecture Overview

The Business Intelligence architecture for this project was designed as a multi-branch data pipeline built entirely within Alteryx Designer. The pipeline follows a structured Extract, Transform, and Load (ETL) approach, where each of the three source datasets is processed independently in its own dedicated container before being integrated into a single enriched output file. This modular design ensures that each branch can be inspected, debugged, and modified independently without affecting the other branches, which is a critical advantage when working with datasets of this scale.

The architecture consists of four logical containers. The first is the Weather Events container, which ingests the raw US weather event records, performs date parsing, airport code standardization, duplicate removal, severity flagging, and daily aggregation. The second is the Flight Delay container, which ingests the 2016 airline performance records, selects and renames the relevant columns, applies null and negative value treatment, and creates binary indicator and segmentation fields. The third is the Airport Metadata container, which ingests the airport reference file, standardizes column names and data types, and removes duplicate airport entries. The fourth and final container is the Final Output container, which joins all three processed branches together using sequential join operations on matching keys, applies a final data cleanse pass, creates the four analytical categorization features, and writes the completed dataset to a CSV output file.

The final output, `Final_Weather_Flight_BI_Dataset.csv`, serves as the single source of truth for all Power BI dashboards in Phase 3. By consolidating flight performance, weather conditions, and airport geographic information into a single flat file, the architecture eliminates the need for any further data preparation at the visualization layer and allows Power BI to focus entirely on delivering interactive analytical insights.

2.0 Data Input from Multiple Datasets

This workflow ingests data from three separate structured CSV datasets, each representing a distinct domain of information relevant to the analysis.

The first input is the US Weather Events dataset, loaded using the Input Data tool reading from `WeatherEvents_Jan2016-Dec2022.csv`. This file contains over 8.6 million records spanning weather events recorded across the United States from January 2016 to December 2022. It includes 14 columns covering event identifiers, weather type, severity, start and end timestamps, precipitation amounts, ICAO airport codes, and geographic coordinates. For this project, only the 2016 records are used, which are isolated using a Filter tool later in the branch.

The second input is the 2016 Airline Delay and Cancellation dataset, loaded using the Input Data tool reading from `2016.csv`. This file contains 5,617,658 domestic US flight records across 28 columns, covering all scheduled commercial flights operated throughout 2016. Key columns include flight date, operating carrier, origin and destination airport codes, scheduled and actual departure and arrival times, delay minutes broken down by cause

(weather, carrier, NAS, security, and late aircraft), cancellation indicator, and cancellation code.

The third input is the US Airport Metadata dataset, loaded using the Input Data tool reading from airports.csv. This file contains 341 records across 7 columns, providing the reference information needed to enrich flight records with geographic and descriptive airport details. Columns include the IATA airport code, full airport name, city, state, country, latitude, and longitude. This dataset acts as a lookup table that is joined to the flight data to add spatial context for geographic mapping in Power BI.

3.0 Data Cleaning

Data cleaning is applied across multiple branches of the workflow to address missing values, duplicate records, and data type inconsistencies identified during initial data exploration.

3.1 Missing Values

In the Flight Delay branch, the Data Cleanse macro (Tool 15) is first applied to all seven delay columns, namely DEP_DELAY, ARR_DELAY, CARRIER_DELAY, WEATHER_DELAY, NAS_DELAY, SECURITY_DELAY, and LATE_AIRCRAFT_DELAY, to handle null values and ensure type consistency. Following the macro, a Formula tool (Tool 16) applies a more precise cleaning rule to each of these fields, creating _CLEAN versions. The logic replaces both null values and negative values with zero. Null values in delay columns indicate that no delay was recorded, typically for cancelled flights where delay metrics are not populated. Negative values represent early departures or arrivals, which are not considered delays and would distort aggregated delay metrics if left untreated. The formula applied is structured as: IF IsNull([DEP_DELAY]) THEN 0 ELSEIF [DEP_DELAY] < 0 THEN 0 ELSE [DEP_DELAY] ENDIF, and this same pattern is applied to all seven delay fields.

In the Final Output branch, a second Data Cleanse macro (Tool 24) is applied after the weather join to handle any null values introduced into the weather fields for records where no weather data was matched. This ensures that fields such as Total_Weather_Events, Total_Precipitation, Severe_Event_Count, Heavy_Event_Count, and Weather_Types do not contain nulls that could cause errors or blank visuals in Power BI.

3.2 Duplicate Records

In the Weather Events branch, the Unique tool (Tool 6) deduplicates all weather event records by the EventId field, ensuring that each weather event is represented exactly once before the aggregation step. Without this step, duplicated event records would inflate the total event counts and precipitation sums produced by the Summarize tool, leading to inaccurate weather metrics in the final dataset.

In the Airport Metadata branch, a second Unique tool (Tool 20) deduplicates the airport records by Airport_IATA. This ensures that each airport code appears exactly once in the

reference table, preventing row multiplication (fan-out) when the airport metadata is joined to the flight records. If an airport code appeared more than once in the reference table, each flight record matching that code would be duplicated for every matching metadata row.

4.0 Data Transformation

Data transformation encompasses column selection and renaming, date and time parsing, type conversion, filtering, formula-based field derivation, and aggregation, all of which are applied across the three processing branches.

4.1 Column Selection, Renaming, and Type Casting

Select tools are applied in all three branches to retain only the columns needed for analysis and to standardize column names across datasets. In the Weather Events branch, the Select tool (Tool 2) retains Weather_Type, Weather_Severity, Weather_StartTime, Weather_EndTime, and Weather_AirportCode, while dropping columns such as TimeZone, LocationLat, LocationLng, County, and ZipCode that are not required for this analysis. In the Flight Delay branch, the Select tool (Tool 14) retains the 12 most relevant flight performance columns, renames them for readability (e.g., FL_DATE to Flight_Date, ORIGIN to Origin_Airport), and explicitly casts all delay columns to Double type to ensure numeric operations can be performed. In the Airport Metadata branch, the Select tool (Tool 19) renames all columns with a consistent prefix convention (e.g., IATA to Airport_IATA, AIRPORT to Airport_Name) and casts LATITUDE and LONGITUDE to Float type for compatibility with Power BI's geographic mapping functions.

4.2 Date Parsing and Standardization

The DateTime tool (Tool 3) in the Weather Events branch converts the Weather_StartTime string field into a proper DateTime data type field named Weather_Date, using the format yyyy-MM-dd hh:mm:ss. This conversion is necessary before any date-based operations can be performed. A subsequent Formula tool (Tool 4) then extracts only the date portion using DateTimeFormat([Weather_Date], "%Y-%m-%d"), producing the Weather_Date_Only field. This standardized date string is used as one half of the composite join key when linking the weather summary to flight records by matching on both airport code and date.

4.3 Filtering

The Filter tool (Tool 5) in the Weather Events branch applies the condition DateTimeYear([Weather_Date]) = 2016 to isolate only the records relevant to this project's scope. This reduces the full 8.6 million record dataset down to approximately 1,185,991 weather event records from the year 2016, significantly reducing processing load for all downstream tools in the branch.

4.4 Airport Code Standardization

A Formula tool (Tool 4) also handles the ICAO to IATA airport code conversion by applying `Right([Weather_AirportCode], 3)` to strip the leading "K" prefix from all weather station codes (e.g., KATL → ATL, KORD → ORD). This standardization is essential because the weather dataset uses ICAO format while both the flight and airport metadata datasets use the 3-letter IATA format. Without this transformation, the join keys would never match and the integration step would produce zero results.

4.5 Aggregation

The Summarize tool (Tool 11) aggregates the weather data from event-level records to a daily airport-level summary. Records are grouped by `Airport_IATA` and `Weather_Date_Only`, producing five summary measures: `Total_Weather_Events` (count of `EventId`), `Total_Precipitation` (sum of `Precipitation_Num`), `Severe_Event_Count` (sum of `Severe_Flag`), `Heavy_Event_Count` (sum of `Heavy_Flag`), and `Weather_Types` (a comma-separated concatenation of all weather types recorded at that airport on that day). This aggregation reduces the 1,185,991 event-level records to a compact daily summary that can be efficiently joined to the 5.6 million flight records on the composite airport-date key.

5.0 Data Integration

Data integration is performed in the Final Output container through two sequential join operations that combine all three processed branches into a single unified dataset.

The first join (Tool 22) connects the processed flight data (left input) with the airport metadata (right input) on the condition that `Origin_Airport` equals `Airport_IATA`. This join enriches every flight record with the corresponding airport name, city, state, latitude, and longitude, enabling geographic analysis and map visualizations in Power BI. An inner join is used, meaning only flights whose origin airport code exists in the airport metadata table are retained. Given that the airport metadata covers 341 primary commercial airports which represent the full scope of US commercial aviation, this does not result in meaningful data loss.

The second join (Tool 23) connects the flight-airport combined dataset (left input) with the aggregated daily weather summary (right input) on a composite key consisting of two conditions: `Origin_Airport` must equal `Airport_IATA`, and `Flight_Date` must equal `Weather_Date_Only`. This links each flight record to the weather conditions that were recorded at its origin airport on the same calendar date, enabling direct analysis of the relationship between weather severity and flight delay or cancellation outcomes. An inner join is used here as well, as discussed in the challenges section.

6.0 Feature Creation

Feature creation is applied in two Formula tools across the workflow, producing a total of nine new derived fields that serve as the primary analytical dimensions and metrics for the Power BI dashboards.

In the Weather Events branch, Formula tool (Tool 7) creates `Severe_Flag` and `Heavy_Flag` as binary integer fields that mark whether each individual event reached a severity level of Severe or Heavy respectively. These flags enable the Summarize tool to count the number of high-severity events per airport per day, producing the `Severe_Event_Count` and `Heavy_Event_Count` summary fields used in the final risk classification.

In the Flight Delay branch, Formula tool (Tool 29) creates five derived fields. `Is_Delayed` is a binary indicator set to 1 if either departure or arrival delay exceeds zero, providing a simple flight-level delay flag for aggregation. `Is_Weather_Delayed` is set to 1 if the weather delay component is greater than zero, isolating flights specifically affected by weather from those delayed by other causes. `Is_Cancelled` converts the numeric cancellation indicator to a binary integer for consistent aggregation. `Flight_Month` extracts the calendar month number from `Flight_Date` to support monthly trend analysis in Power BI. `Route` concatenates `Origin_Airport` and `Destination_Airport` into a human-readable string such as "ATL to ORD" for route-level filtering.

In the Final Output branch, Formula tool (Tool 25) creates four categorical segmentation features. `Delay_Category` classifies each flight into one of four bands based on `DEP_DELAY`: "No Delay" for zero minutes, "Minor Delay" for up to 15 minutes, "Moderate Delay" for up to 60 minutes, and "Severe Delay" for anything beyond 60 minutes. `Weather_Risk_Level` classifies each flight's airport-day context into "High Risk" if any Severe or Heavy weather event was recorded, "Medium Risk" if three or more total events occurred, "Low Risk" if one or two events occurred, and "No Weather Risk" if no weather events were recorded. `Cancellation_Status` provides a readable label of "Cancelled" or "Completed" for dashboard filtering. `Weather_Delay_Status` labels each flight as "Weather Delayed" or "Not Weather Delayed" for direct comparison of weather-affected versus unaffected flights.

7.0 Analytical Techniques Applied

7.1 Trend Analysis

Monthly trend analysis is supported through the `Flight_Month` derived field, which enables Power BI to aggregate total weather delay minutes, total weather events, cancellation counts, and delay counts by month across the full 2016 calendar year. This allows the identification of seasonal patterns in weather disruption, such as elevated delays in winter months due to snow and ice events or in summer months due to storm activity.

7.2 Segmentation

Segmentation is implemented through the four categorical features created in the Final Output branch. The `Delay_Category` field segments all flights into severity bands that allow

comparison of delay magnitude distribution across airports, carriers, and weather conditions. The Weather_Risk_Level field segments each airport-day into a four-tier risk classification based on the quantity and severity of weather events, enabling identification of high-risk airports and time periods. Together, these segmentation fields form the primary filter dimensions in the Power BI dashboards, allowing users to interactively slice the data by delay severity, weather risk, cancellation status, and weather delay attribution.

7.3 Correlation and Comparative Analysis

The composite join between flight records and daily weather summaries at the airport level creates the data structure necessary for correlation analysis between weather event frequency or severity and flight delay or cancellation outcomes. By pairing each flight with the total number of weather events, precipitation levels, and high-severity event counts at its origin airport on the same day, the dataset enables Power BI scatter plots and bar charts to reveal whether airports with more frequent or more severe weather events also exhibit higher average delay minutes or cancellation rates.

8.0 Alteryx Workflow

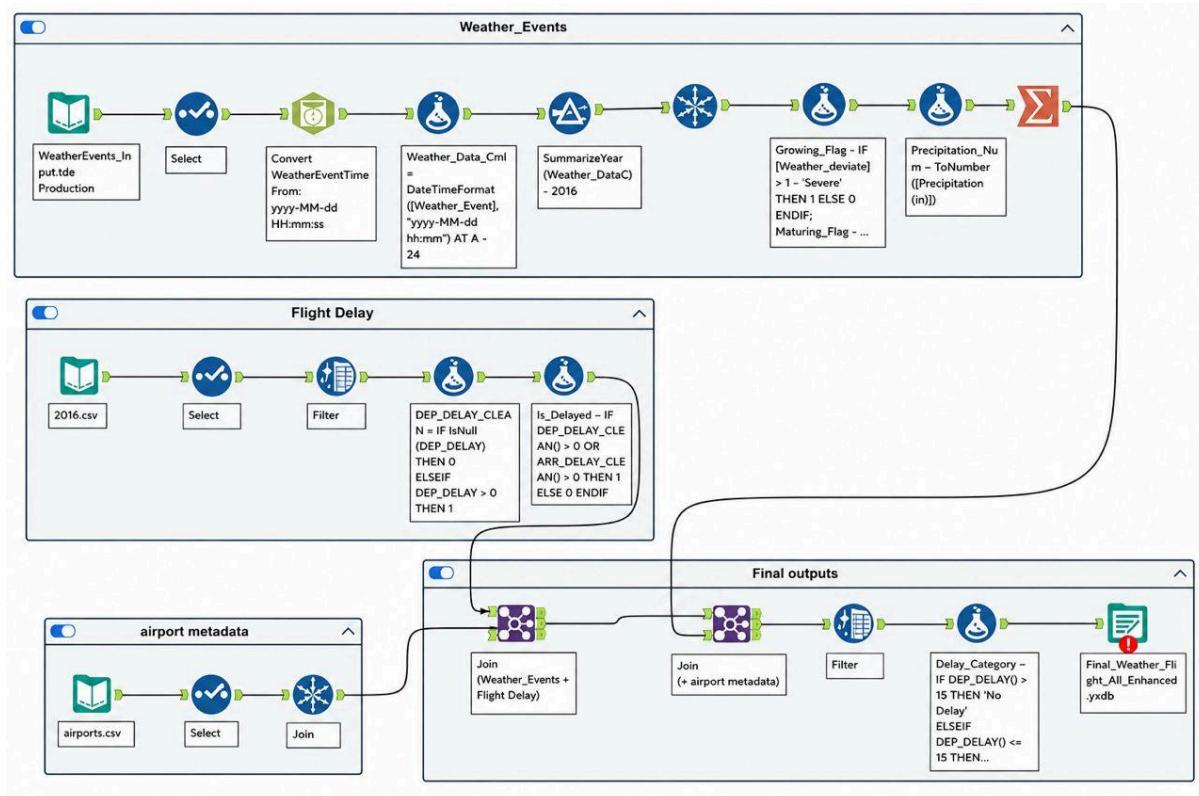


Figure 8.1: Complete Alteryx workflow used for data integration, cleansing, transformation, and preparation of flight delay, airport metadata, and weather datasets

The Alteryx workflow shown in Figure 8.1 implements a complete Extract, Transform, and Load (ETL) pipeline for integrating flight delay, airport metadata, and weather event datasets into a unified analytical dataset for Power BI visualization. The workflow is organized into multiple logical branches, each responsible for processing a specific source dataset independently before integration.

The Weather Events branch performs data filtering, date parsing, airport code standardization, duplicate removal, severity flag creation, and daily weather aggregation. The Flight Delay branch applies column selection, data type conversion, null and negative value treatment, and creation of derived analytical features such as delay indicators and route information. The Airport Metadata branch standardizes airport reference data and removes duplicate airport entries to ensure accurate joins.

The processed datasets are integrated in the Final Output branch through sequential join operations based on airport and date matching keys. A final cleansing and transformation stage is then applied to generate categorical analytical fields including delay severity, weather risk level, cancellation status, and weather delay classification. The completed dataset is exported as a CSV file and serves as the centralized data source for subsequent Power BI dashboard development and analysis.

9.0 Tools Used Summary

Tool	Category	Purpose in Workflow
Input Data	In/Out	Load all three raw CSV source files
Select	Preparation	Column selection, renaming, and type casting across all three branches
DateTime	Parse	Convert weather timestamp string to proper DateTime type
Formula	Transform	Date formatting, ICAO-to-IATA conversion, null/negative cleaning, binary flags, segmentation categories, route creation
Filter	Preparation	Isolate 2016 weather records from the full 2016–2022 dataset
Unique	Preparation	Remove duplicate EventId records in weather branch and duplicate Airport_IATA records in airport branch
Summarize	Transform	Aggregate weather events to daily airport-level summary with counts, sums, and concatenated types
Data Cleanse (Macro)	Preparation	Handle null values in delay columns and post-join weather fields

Join	Join	Two sequential inner joins to integrate flight, airport, and weather data on matching keys
Output Data	In/Out	Write the final enriched dataset to CSV for Power BI ingestion

10.0 Challenges Faced

Challenge 1: ICAO to IATA Airport Code Mismatch

The weather dataset uses ICAO airport codes following the US convention of a leading "K" prefix (e.g., KATL for Atlanta Hartsfield, KORD for Chicago O'Hare), while both the flight performance dataset and the airport metadata dataset use the standard 3-letter IATA format (e.g., ATL, ORD). When the initial join was attempted without any code standardization, the result was zero matched records because no weather ICAO code would ever directly match an IATA code. This was resolved by applying the expression `Right([Weather_AirportCode], 3)` in the Formula tool to extract the last three characters of every ICAO code, effectively stripping the K prefix and producing the equivalent IATA code. This approach is reliable for all standard US commercial airport codes which consistently follow the four-character KXXX format.

Challenge 2: Join Type Selection — Left Join versus Inner Join

The data integration plan documented in Phase 1 specified using a left join for the weather-to-flight connection in order to retain all flight records regardless of whether weather data was available at that airport on a given day. During implementation, applying a left join produced a very large number of records with null values across all weather fields, representing flights at airports or on dates where no weather events were recorded in the weather dataset. The presence of these null-heavy records reduced the quality and interpretability of analytical outputs in Power BI, as aggregations involving weather fields became unreliable due to the high proportion of missing values. An inner join was therefore applied instead, ensuring that every record in the final dataset has complete and meaningful values across all flight, airport, and weather fields. This deviation from the original plan means that the analysis reflects only flights where weather data was available at the origin airport, which is noted as a scope limitation that will be formally addressed and reconciled in the Phase 4 consolidated report.

Challenge 3: Missing and Negative Delay Values

The raw flight dataset contained a significant number of null values across all delay columns, particularly for cancelled flights where delay metrics are not populated by the data source. In addition, a number of records contained negative delay values representing early departures

or arrivals, which are operationally irrelevant to delay analysis but would negatively affect mean and sum calculations if retained. A two-step cleaning approach was implemented to address both issues. The Data Cleanse macro was first applied to handle null value replacement at the field level. A custom Formula tool was then applied to enforce the rule that any remaining null or negative value should be replaced with zero across all seven delay columns, producing clean non-negative numeric fields for all downstream aggregation and categorization.

Challenge 4: Data Type Inconsistencies Across CSV Inputs

All three source datasets were ingested as flat CSV files, which caused Alteryx to default all fields to string (V_String) data types regardless of their intended numeric or date types. This affected delay columns in the flight dataset, coordinate fields in the airport dataset, and the precipitation field in the weather dataset. Without explicit type conversion, none of these fields could be used in arithmetic operations, geographic mapping, or date-based filtering. This was resolved by using the Select tool to cast delay columns to Double and coordinate columns to Float, and by using the Formula tool to apply `ToNumber()` for precipitation conversion. One minor inconsistency remains in the workflow where `NAS_DELAY_CLEAN` was inadvertently assigned the `V_WString` type instead of Double during formula configuration. This does not cause a workflow error but represents a data type inconsistency that would be corrected in a future revision of the workflow.